Steps to Derive Equations of Motion

1. Define Generalized Coordinates

Identify the degrees of freedom and choose the appropriate generalized coordinate(s) q. - For this problem: $q = \theta$

2. Identify Kinetic and Potential Energy

- $K =$ Kinetic Energy
- $P =$ Potential Energy

3. Form the Lagrangian

- $L = K - P$

4. Identify Generalized Forces (Q)

Generalized forces include non-conservative forces like damping and external torques. - $Q = \tau - b\dot{\theta}$ - τ is the input torque. - $-b\dot{\theta}$ is the damping torque (viscous friction).

5. Apply Euler-Lagrange Equation

For each generalized coordinate q_i:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = Q_i$$

Detailed Calculation for θ:

1. Compute $\frac{\partial L}{\partial \dot{\theta}}$:
   - $\frac{\partial L}{\partial \dot{\theta}} = m\ell^2\dot{\theta}$
2. Compute the time derivative $\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right)$:
   - $\frac{d}{dt}(m\ell^2\dot{\theta}) = m\ell^2\ddot{\theta}$
3. Compute $\frac{\partial L}{\partial \theta}$:
   - $\frac{\partial L}{\partial \theta} = -(mg\ell\cos(\theta) + k_1\theta + k_2\theta^3)$
4. Combine into Euler-Lagrange:
   - $m\ell^2\ddot{\theta} - [-(mg\ell\cos(\theta) + k_1\theta + k_2\theta^3)] = \tau - b\dot{\theta}$
   - $m\ell^2\ddot{\theta} + mg\ell\cos(\theta) + k_1\theta + k_2\theta^3 = \tau - b\dot{\theta}$

6. Solve for $\ddot{\theta}$

Rearrange the equation to solve for the highest derivative:

$$\ddot{\theta} = \frac{1}{m\ell^2}\left(\tau - b\dot{\theta} - mg\ell\cos(\theta) - k_1\theta - k_2\theta^3\right)$$